

FEDERAL UNIVERSITY LOKOJA
DEPARTMENT OF PHYSICS
FIRST SEMESTER EXAMINATION 2022/23 SESSION
PHY 803: QUANTUM MECHANICS
Time: 3 Hrs October, 2024 3 CREDITS.

*Instruction: Answer any **Four** questions.*

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}, \quad \int_{-\infty}^{\infty} x^{(2n)} e^{-\alpha x^2} dx = \frac{1 \times 3 \times 5 \times 7 \times \cdots \times (2n-1)}{2^n \alpha^n} \sqrt{\frac{\pi}{\alpha}}$$
$$\int_0^{\infty} x^{(2n+1)} e^{-\alpha x^2} dx = \frac{n!}{2\alpha^{n+1}}, \quad \int_0^{\infty} x^n e^{-x} dx = n!$$

1. (a) State Heisenberg's uncertainty principle.
(b) Using an arbitrary wave function where necessary derive the commutation relations between (i) the position and momentum components $\hat{X}$ and $\hat{P}_x$, (ii) the angular momentum components $\hat{L}_x$ and $\hat{L}_y$.
(c) Obtain the generalized uncertainty principle between two non commuting linear operators $\hat{A}$ and $\hat{B}$. Use your results in b(i) and b(ii) to verify the uncertainty relation between $\hat{X}$ and $\hat{P}_x$ and, $\hat{L}_x$ and $\hat{L}_y$.
2. (a) If an orthonormal basis set $\{|\phi_i\rangle\}$ belongs to the same space as an arbitrary ket $|\omega\rangle = \sum_i c_i |\phi_i\rangle$. Using the Dirac notation, obtain the completeness relations for (i) the expansion coefficient c_i , (ii) the set $\{|\phi_i\rangle\}$.
(b) Show that the eigenvalues of an Hermitian operator are real and the eigenvectors corresponding to different eigenvalues are orthogonal.
(c) The state space of a certain physical system is 3-dimensional. If $\{|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle\}$ is an orthonormal basis of the space. Given the kets $|\psi_1\rangle = \sqrt{2}|\phi_1\rangle + i|\phi_2\rangle + |\phi_3\rangle$, $|\psi_2\rangle = |\phi_1\rangle + i|\phi_3\rangle$,
i) Normalize the kets $|\psi_1\rangle$ and $|\psi_2\rangle$.
ii) Determine the matrices ρ_1 and ρ_2 representing the projection operators onto $|\psi_1\rangle$ and $|\psi_2\rangle$ respectively. Verify the hermiticity of the projection operators.
3. (a) Why do we consider perturbation theory as an important theory?
(b) Show that the first order correction to the energy of a time-independent perturbed system is the expectation value of the perturbation in the unperturbed state.
(c) Calculate the first order correction to the ground state energy of a simple harmonic oscillator perturbed by αx^4 , where α is a constant and the ground state wavefunction is $\phi_o = (m\omega/\pi\hbar)^{1/4} \exp(-m\omega x^2/2\hbar)$
4. (a) Distinguish between the intrinsic spin and orbital angular momentum of a particle.
(b) i) List the quantization of direction for the total angular momentum $\vec{J}$.
ii) Calculate the allowed directions θ for momentum $j = 1$.
iii) Find the energy levels of a system with an intrinsic spin $s = \frac{3}{2}$ whose Hamiltonian is give by $\hat{H} = \frac{1}{\hbar^2} [\hat{S}_x^2 + \hat{S}_y^2 - 2\hat{S}_z^2] - \frac{3}{\hbar} \hat{S}_z$.

- (c) For a system of total angular momentum $j = 1$, the operators $\hat{J}_x$, $\hat{J}_y$ and $\hat{J}_z$ are given by

$$\hat{J}_x = \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \hat{J}_y = \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{bmatrix} \quad \hat{J}_z = \hbar \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

- i) What values of $\hat{J}_x$ will be obtained upon measurement?
 ii) Take the state in which $j_x = -\hbar$. Calculate $\langle \hat{J}_z \rangle$, $\langle \hat{J}_z^2 \rangle$ and the uncertainty $\Delta \hat{J}_z$.

5. (a) What is a stationary state of quantum system? State the properties of a stationary state.
 (b) The Hamiltonian $\hat{H}$ and the initial state $|\psi_i\rangle$ of a physical system are respectively given by

$$\hat{H} = \varepsilon \begin{bmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad |\psi_i\rangle = \frac{1}{\sqrt{3}} \begin{bmatrix} i \\ 1-i \\ 1 \end{bmatrix}$$

where ε has the dimension of energy. What are the energy eigenvalues and their respective probabilities.

- (c) A free particle in one dimension ($H = p^2/2m$) is in the state $|1\rangle$ whose normalized wave function is $\psi_1(x) = (\frac{\alpha}{\pi})^{1/4} \exp(-\alpha x^2/2)$. Calculate the mean values of p^2 . Hence determine the uncertainty in p^2 if $\langle p^4 \rangle = \frac{3}{4} \alpha^2 \hbar^4$.
6. (a) State Pauli's exclusion principle. Use an appropriate wavefunction for two body system to demonstrate the principle.
 (b) i) What is an exchange operator? Deduce the eigenvalues for the exchange operator $\hat{P}_{ij}$.
 ii) Distinguish between symmetric and antisymmetric wavefunctions.
 (c) Given the lowest member of states of the Hydrogen atom $\psi_{100} = 2a_o^{-3/2} \exp(-(r/a_o))$ and $\psi_{210} = (24a_o^5)^{-1/2} r \exp(-(r/2a_o))$.
 i) Calculate the value of r at which the radial probability density of the Hydrogen atom reaches its maximum for the $n = 2, l = 1$ state.
 ii) Compute the expectation values of the $\langle r \rangle$ and $\langle r^2 \rangle$ for the $n = 2, l = 1$ state.
 iii) Determine the width Δr for the state.
 iv) Compare your results in c(i) and c(ii).